Tutorial 8 (Gear)

- Problem 9-1
- Problem 9-5
- Problem 9-7
- Problem 9-25 (use the data from Row 1 of the table)
- Problem 9-35 (ignore the part on efficiency)
- Problem 9-39

*†9-1 A 24-tooth gear has AGMA standard full-depth involute teeth with diametral pitch of 5. Calculate the pitch diameter, circular pitch, addendum, dedendum, tooth thickness, and clearance.

*9-5 A spur gearset must have pitch diameters of 2.5 and 8 in. What is the largest standard tooth size, in terms of diametral pitch p_d, that can be used without having any interference or undercutting? Find the number of teeth on the hob-cut gear and pinion for this p_d:

a. For a 20° pressure angle.

b. For a 25° pressure angle. (Note that diametral pitch need not be an integer.)

*†9-7 Design a simple, spur gear train for a ratio of +6:1 and diametral pitch of 5. Specify pitch diameters and numbers of teeth. Calculate the contact ratio.

*†9-25 Figure P9-1 shows a compound planetary gear train (not to scale). Table P9-1 gives data for gear numbers of teeth and input velocities. For the row(s) assigned, find the variable represented by a question mark.

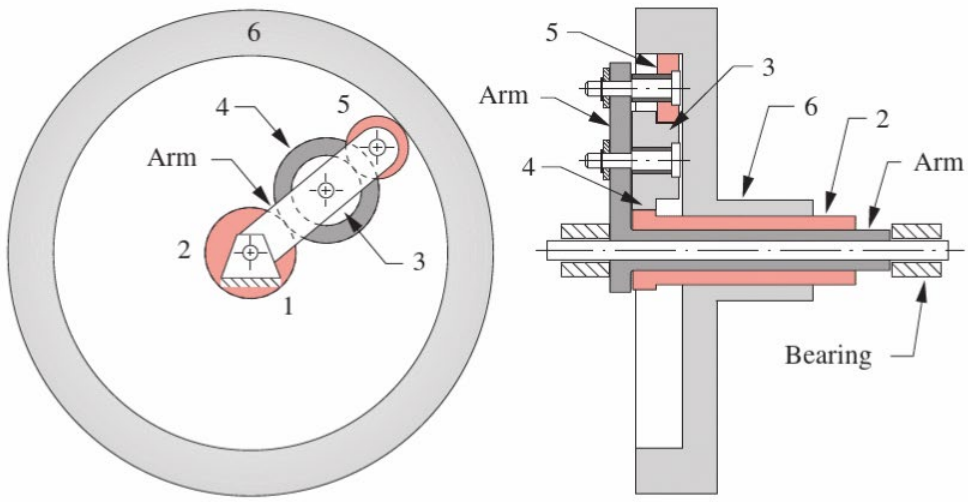


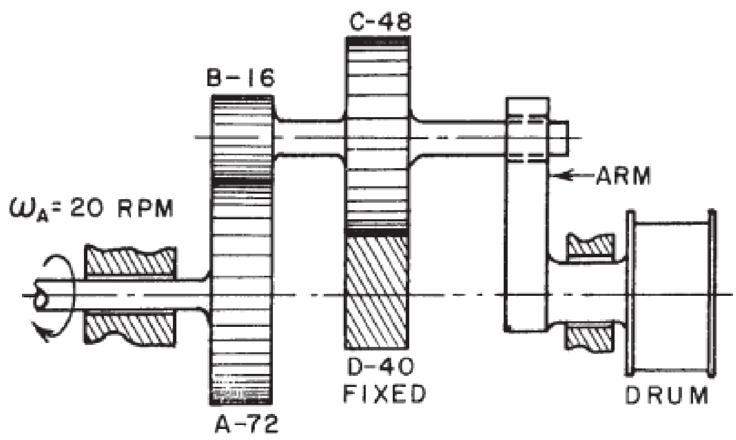
FIGURE P9-1

Planetary gearset for Problem 9-25 and 9-81

TABLE P9-1 Data for Problem 9-25 and 9-81

Row	N_2	N_3	N_4	N_5	N_6	ω_2	ω_6	ω_{arm}
a	30	25	45	50	200	?	20	– 50
b	30	25	45	50	200	30	?	– 90
c	30	25	45	50	200	50	0	?
d	30	25	45	30	160	?	40	– 50
e	30	25	45	30	160	50	?	– 75
f	30	25	45	30	160	50	0	?

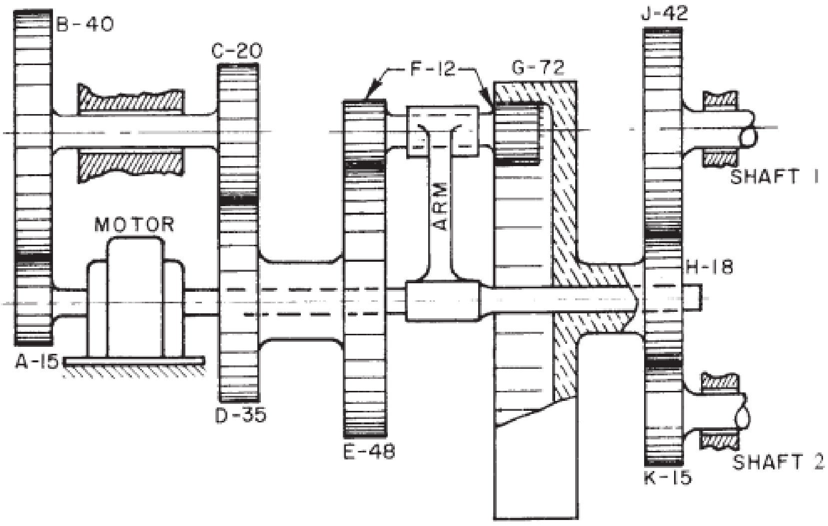
*†9-35 Figure P9-5a shows a compound epicyclic train used to drive a winch drum. Gear *A* is driven at 18 rpm CW and gear *D* is fixed to ground. Tooth numbers are in the figure. Find speed and direction of the drum. What is train efficiency for gearsets $E_0 = 0.97$?



(a)

FIGURE P9-5

*†9-39 Figure P9-7a shows a gear train containing both compound-reverted and epicyclic stages. Tooth numbers are in the figure. The motor is driven CW at 1500 rpm. Find the speeds of shafts 1 and 2.



B-40
C-20
F-12
G-72
J-42
SHAFT 1
MOTOR
ARM
H-18
A-15
D-35
E-48
K-15
SHAFT 2

(*a*)

FIGURE P9-7